

SIMATS ENGINEERING

Department of Computer Science & Engineering

CO5-ClassActivity

Programming in Java - File I/O Operations

Student Name	K. V. S. S. Jeswanth Babu
Register No.	192472012
Course Code	CSA0904
Course Name	Programming in Java
Assignment Topic	CO5-ClassActivity: Count Characters and Lines
Question No.	Question 4

QUESTION 4: COUNT LINES, WORDS, AND CHARACTERS IN FILE

Problem Statement:

Create a Java program to read a text file (source.txt) and compute statistical metrics including total number of lines, total words, total characters, and character counts without whitespace.

Source Code (CountFileStats.java):

```
import java.io.BufferedReader;
import java.io.File;
import java.io.FileReader;
import java.io.IOException;

public class CountFileStats {
    public static void main(String[] args) {
        String fileName = "source.txt";
        File file = new File(fileName);

        if (!file.exists()) {
            System.out.println("Error: The file '" + fileName + "' does not exist.");
            return;
        }

        int lineCount = 0;
        int charCountWithSpaces = 0;
        int charCountNoSpaces = 0;
        int wordCount = 0;

        try (BufferedReader reader = new BufferedReader(new FileReader(file))) {
            String line;

            while ((line = reader.readLine()) != null) {
                lineCount++;
                charCountWithSpaces += line.length();
                charCountNoSpaces += line.replaceAll("\\s+", "").length();

                String trimmed = line.trim();
                if (!trimmed.isEmpty()) {
                    String[] words = trimmed.split("\\s+");
                    wordCount += words.length;
                }
            }

            System.out.println("=====");
            System.out.println("          FILE STATISTICS SUMMARY          ");
            System.out.println("=====");
            System.out.println("File Name           : " + file.getName());
            System.out.println("Total Lines         : " + lineCount);
            System.out.println("Total Words          : " + wordCount);
            System.out.println("Total Characters (Raw) : " + charCountWithSpaces);
            System.out.println("Characters (No Spaces) : " + charCountNoSpaces);
            System.out.println("=====");

        } catch (IOException e) {
            System.err.println("Error reading the file: " + e.getMessage());
            e.printStackTrace();
        }
    }
}
```

EXECUTION & VERIFICATION OUTPUT

1. Program Execution & Computed Statistics:

The screenshot shows an IDE with a project named CSA0904. The Explorer panel on the left lists files including CO1-AT1, CO1-AT2, CO4-AT1, CO4-AT2, CO5-ClassActivity, CopyFileContent.java, CountFileStats.java, CreateAndWriteFile..., destination.txt, output.txt, ReadAndDisplayFile..., RecreateFile.java, sample_file.txt, and source.txt. The source.txt file is selected and its content is displayed in the editor:

```
1 Hello, this is the source file content.
2 Learning Java File I/O operations:
3 1. File Creation
4 2. Reading from a file
5 3. Writing to a file
6 4. Copying content between files
7 File copy operation test successful!
```

The Terminal panel at the bottom shows the command `java CountFileStats.java` being executed. The output displays the file statistics summary:

```
=====
FILE STATISTICS SUMMARY
=====
File Name       : source.txt
Total Lines     : 7
Total Words     : 35
Total Characters (Raw) : 199
Characters (No Spaces) : 171
=====
```

2. Source File Content Verification (source.txt):

The screenshot shows the same IDE with the CountFileStats.java file selected in the editor. The code is as follows:

```
1 import java.io.BufferedReader;
2 import java.io.File;
3 import java.io.FileReader;
4 import java.io.IOException;
5
6 public class CountFileStats {
7     Run | Debug
8     public static void main(String[] args) {
9         String fileName = "source.txt";
10        File file = new File(fileName);
11
12        if (!file.exists()) {
13            System.out.println("Error: The file '" + fileName + "' does not exist.");
14            return;
15        }
16
17        int lineCount = 0;
18        int charCountWithSpaces = 0;
19        int charCountNoSpaces = 0;
20        int wordCount = 0;
21
22        try (BufferedReader reader = new BufferedReader(new FileReader(file))) {
```

The Terminal panel shows the command `java countFileStats.java` being executed. The output displays the file statistics summary:

```
=====
FILE STATISTICS SUMMARY
=====
File Name       : source.txt
Total Lines     : 7
Total Words     : 35
Total Characters (Raw) : 199
Characters (No Spaces) : 171
=====
```